# RPractical_Paciente

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#Part A1

```
v<-seq(from=5, to=50, by=5)
length(v)
```

```
## [1] 10
```

```
sum(v)
```

```
## [1] 275
```

```
mean(v)
```

```
## [1] 27.5
```

#Part A2

```
w<-rep(c(1,2,3),4)
z<-rep(c(1,2,3), each=4)
w
```

```
##  [1] 1 2 3 1 2 3 1 2 3 1 2 3
```

```
z
```

```
##  [1] 1 1 1 1 2 2 2 2 3 3 3 3
```

#Part A3

```
17 %% 5
```

```
## [1] 2
```

```
17 %/%5
```

```
## [1] 3
```

```r
2^10
```

```
## [1] 1024
```

```r
sqrt(144)
```

```
## [1] 12
```

```r
round(pi, 3)
```

```
## [1] 3.142
```

#Part A4

```r
a<-c(2,4,6,8)
b<-c(1,3,5,7)
a+b
```

```
## [1]  3  7 11 15
```

```r
a*b
```

```
## [1]  2 12 30 56
```

```r
a>b
```

```
## [1] TRUE TRUE TRUE TRUE
```

```r
sum(a*b)
```

```
## [1] 100
```

#Part A5

```r
paste("Student", 1:3, sep = "-")
```

```
## [1] "Student-1" "Student-2" "Student-3"
```

```r
typeof(42)
```

```
## [1] "double"
```

```r
typeof(42L)
```

```
## [1] "integer"
```

```r
typeof("42")
```

```
## [1] "character"
```

```r
typeof(41i)
```

```
## [1] "complex"
```

```r
typeof(TRUE)
```

```
## [1] "logical"
```

#Part B1

```r
students<-data.frame(
  id = 1:8,
  name = c("Alice", "Bruno", "Chen", "Dana", "Elif", "Farid", "Grace", "Hiro"),
  sex = c("F", "M", "M", "F", "F", "M", "F", "M"),
  midterm = c(78, 85, 92, 67, 74, 88, 95, 89),
  final = c(82,79,96,71,80,91,93,65),
  attendance = c(95,88,100,72,90,85,98,60),
  stringAsFactors = FALSE
)
students
```

```
##   id  name sex midterm final attendance stringAsFactors
## 1  1 Alice   F      78    82         95           FALSE
## 2  2 Bruno   M      85    79         88           FALSE
## 3  3  Chen   M      92    96        100           FALSE
## 4  4  Dana   F      67    71         72           FALSE
## 5  5  Elif   F      74    80         90           FALSE
## 6  6 Farid   M      88    91         85           FALSE
## 7  7 Grace   F      95    93         98           FALSE
## 8  8  Hiro   M      89    65         60           FALSE
```

```r
#Part B2
str(students)
```

```
## 'data.frame':    8 obs. of  7 variables:
##  $ id            : int  1 2 3 4 5 6 7 8
##  $ name          : chr  "Alice" "Bruno" "Chen" "Dana" ...
##  $ sex           : chr  "F" "M" "M" "F" ...
##  $ midterm       : num  78 85 92 67 74 88 95 89
##  $ final         : num  82 79 96 71 80 91 93 65
##  $ attendance    : num  95 88 100 72 90 85 98 60
##  $ stringAsFactors: logi  FALSE FALSE FALSE FALSE FALSE FALSE ...
```

```r
dim(students)
```

```
## [1] 8 7
```

```r
colnames(students)
```

```
## [1] "id"              "name"            "sex"             "midterm"
## [5] "final"           "attendance"      "stringAsFactors"
```

```r
summary(students)
```

```
##        id              name          sex       midterm          final
##  Min.   :1.00   Length   :8   Length   :8   Min.   :67.00   Min.   :65.00
##  1st Qu.:2.75   N.unique :8   N.unique :2   1st Qu.:77.00   1st Qu.:77.00
##  Median :4.50   N.blank  :0   N.blank  :0   Median :86.50   Median :81.00
##  Mean   :4.50   Min.nchar:4   Min.nchar:1   Mean   :83.50   Mean   :82.12
##  3rd Qu.:6.25   Max.nchar:5   Max.nchar:1   3rd Qu.:89.75   3rd Qu.:91.50
##  Max.   :8.00                               Max.   :95.00   Max.   :96.00
##    attendance     stringAsFactors
##  Min.   : 60.00   Mode :logical
##  1st Qu.: 81.75   FALSE:8
##  Median : 89.00
##  Mean   : 86.00
##  3rd Qu.: 95.75
##  Max.   :100.00
```

```r
#Part B3
students$average<-round((students$midterm + students$final)/ 2, 1)
students
```

```
##   id  name sex midterm final attendance stringAsFactors average
## 1  1 Alice   F      78    82         95           FALSE    80.0
## 2  2 Bruno   M      85    79         88           FALSE    82.0
## 3  3  Chen   M      92    96        100           FALSE    94.0
## 4  4  Dana   F      67    71         72           FALSE    69.0
## 5  5  Elif   F      74    80         90           FALSE    77.0
## 6  6 Farid   M      88    91         85           FALSE    89.5
## 7  7 Grace   F      95    93         98           FALSE    94.0
## 8  8  Hiro   M      89    65         60           FALSE    77.0
```

```r
#Part B4
students$passed<-students$average>=70 & students$attendance>= 75
students
```

```
##   id  name sex midterm final attendance stringAsFactors average passed
## 1  1 Alice   F      78    82         95           FALSE    80.0   TRUE
## 2  2 Bruno   M      85    79         88           FALSE    82.0   TRUE
## 3  3  Chen   M      92    96        100           FALSE    94.0   TRUE
## 4  4  Dana   F      67    71         72           FALSE    69.0  FALSE
## 5  5  Elif   F      74    80         90           FALSE    77.0   TRUE
## 6  6 Farid   M      88    91         85           FALSE    89.5   TRUE
## 7  7 Grace   F      95    93         98           FALSE    94.0   TRUE
## 8  8  Hiro   M      89    65         60           FALSE    77.0  FALSE
```

```r
#Part B5
students$standing<-
  ifelse(students$average>=90, "Honors",
         ifelse(students$average>=75, "Good", "Needs support"))
students
```

```
##   id  name sex midterm final attendance stringAsFactors average passed
## 1  1 Alice   F      78    82         95           FALSE    80.0   TRUE
## 2  2 Bruno   M      85    79         88           FALSE    82.0   TRUE
## 3  3  Chen   M      92    96        100           FALSE    94.0   TRUE
## 4  4  Dana   F      67    71         72           FALSE    69.0  FALSE
## 5  5  Elif   F      74    80         90           FALSE    77.0   TRUE
## 6  6 Farid   M      88    91         85           FALSE    89.5   TRUE
## 7  7 Grace   F      95    93         98           FALSE    94.0   TRUE
## 8  8  Hiro   M      89    65         60           FALSE    77.0  FALSE
##        standing
## 1          Good
## 2          Good
## 3        Honors
## 4 Needs support
## 5          Good
## 6          Good
## 7        Honors
## 8          Good
```

```r
#Part B6
classify <- function(score) {

}
```